TEST 2 — ONE-PAGE NOTES

Copy by hand onto ONE SIDE of A4. No drawings. This typed reference is not an admissible exam sheet.

FOUNDATIONS (~2 questions)

Arthropoda: jointed limbs, segmented/tagmatised body, chitin exoskeleton, ecdysis (moulting). Panarthropoda = Arthropoda + Onychophora + Tardigrada; Ecdysozoa includes nematodes. Pancrustacea = crustaceans + hexapods; “Crustacea” excluding insects is paraphyletic. Monophyletic = ancestor + all descendants; paraphyletic omits descendants.

Insects: head/thorax/abdomen; 1 antennal pair; 3 leg pairs; wings on meso-/metathorax. Crustaceans usually 2 antennal pairs, biramous limbs. Myriapods: centipedes 1 leg-pair/segment, millipedes 2/diplosegment.

Holometabola: egg–larva–pupa–adult; separate niches reduce larval/adult competition. Hemimetabola: egg–nymph–adult, no pupa. Exarate = free pupal appendages; obtect = attached; coarctate = within puparium; decticous = functional mandibles.

Labels: antenna scape–pedicel–flagellum; leg coxa–trochanter–femur–tibia–tarsus–pretarsus. Tergite dorsal, sternite ventral; pronotum = prothoracic dorsal plate. Spiracles/tracheae deliver O2; Malpighian tubules excrete; open circulation.

INSECT OVERVIEW (~2)

Hemiptera: piercing stylets in labial rostrum; incomplete metamorphosis; true bugs often hemelytra (hard base, membrane tip). Beetles chew; elytra hardened throughout. Sap-feeders: honeydew, filter chamber, plant-disease vectors.

Odonata: predatory naiads, grasping labium. Mayflies: aquatic naiads, winged subimago. Stoneflies: aquatic nymphs, 2 cerci. Neoptera fold wings; palaeopterans cannot. Orthoptera jumping hindlegs; earwigs forceps cerci; thrips fringed wings; termites nested in cockroaches.

Neuroptera larvae: sucking jaws. Mecoptera: rostrum with chewing tip. Fleas: wingless, laterally compressed jumping adults; legless larvae. Lice ectoparasitic; stick insects camouflage.

COLEOPTERA (~4)

Elytra protect folded hindwings/abdomen, reduce water loss; mobile prothorax; chewing larvae/adults. Diversity: ancient origin, protected flexible body, holometaboly, herbivore radiation with flowering plants.

Rostrum drills oviposition hole (weevils); ambrosia beetles farm fungi. Predators use extraoral digestion; aquatic beetles use air stores/plastrons/gills. Gyrinidae: divided eyes; Hydrophilidae: clubbed antennae collect air, long maxillary palps.

ID clues: Scarabaeidae lamellate closing antennal club; Curculionidae snout + elbowed club; Staphylinidae short elytra; Elateridae prosternal click; Cerambycidae long antennae, notched eyes; Chrysomelidae usually shorter antennae. Tarsal formula = fore/mid/hind; pseudotetramerous = 5 looking like 4.

Dung beetles recycle dung, suppress bush flies; ladybirds biocontrol; dermestids damage collections/stores; wood borers/leaf beetles/weevils pests. Bioluminescence signals mates; neoteny = sexually mature juvenile-like form; Structural colour: light interference; pigments absorb wavelengths.

HYMENOPTERA (~4)

Usually 2 membranous wing pairs joined by hamuli; holometabolous. Symphyta: broad waist, paraphyletic; Apocrita: narrow waist. Propodeum = abdominal segment 1; mesosoma = thorax + propodeum; metasoma = remaining abdomen.

Haplodiploidy: unfertilised eggs usually haploid males; fertilised eggs diploid females. Sting = modified ovipositor, females only. Aculeata stinging group; “Parasitica” not a natural clade. Bees derived from predatory wasps; larvae eat pollen.

Parasite exploits host; parasitoid eventually kills host; predator consumes multiple prey. Idiobiont arrests host development (often external); koinobiont allows development (often internal, narrower host range). Host immunity selects specialisation; some wasps inject immunosuppressing viruses.

Superparasitism: same parasitoid species reuses host; multiparasitism: different species; hyperparasitism: parasitoid attacks parasitoid, can undermine biocontrol. Herbivore-damaged plants release attractants; parasitoids learn host-associated odours.

Eusociality = cooperative brood care + reproductive division of labour + overlapping generations; evolved repeatedly. Subsocial: parental care; parasocial: same-generation females share nest. Mass provisioning = food supplied upfront; progressive = repeated feeding.

Prac: torulus antennal socket; scutellum behind mesoscutum; stigma thickened wing patch; closed cell bounded by veins; calcar tibial spur, basitarsus first tarsomere. Check wing cells, pronotum, antennae, tibial spurs. Ants elbowed antennae/petiole; bees branched hairs.

Benefits: pollination, parasitoid biocontrol. Risks: invasive bees compete/spread disease; social wasps sting/predate bees; Sirex noctilio damages pine. Morphology and host use determine whether beneficial or pest.

AMPHIESMENOPTERA / LEPIDOPTERA (~4)

Trichoptera + Lepidoptera share larval silk spinneret. Caddisflies: hairy wings, haustellum; mostly freshwater larvae, cases/nets, terminal hooks; predators/detritivores/filter-feeders.

Lepidoptera: scales; proboscis from paired maxillary galeae; caterpillars 3 thoracic leg-pairs + abdominal prolegs with crochets. Sawfly larvae lack crochets. Frenulum hindwing bristle hooks forewing retinaculum. Ditrysia: separate mating/egg-laying openings linked internally.

ID: Gelechioidea upturned labial palps, scaled proboscis base. Pyraloidea scaled base + ventral abdominal tympana. Geometroidea naked proboscis + lateroventral abdominal tympana + chaetosemata. Noctuoidea thoracic tympana, naked proboscis. Tortricidae ocelli/chaetosemata; Tineidae woolly head, long maxillary palps. Exceptions occur.

Chaetosemata = sensory scale patches; ocelli = simple photoreceptors. Butterflies nested within moths; “micro/macro” not natural groups. Host-plant arms race: mono-/oligo-/polyphagy = one/few/many hosts. Concealed feeding protects; exposed larvae use camouflage, toxins, warning colours, irritating hairs.

Bat defence: hearing/evasion, sonar jamming, acoustic warning/mimicry; hearing may predate bats. Benefits: pollination, litter recycling, Bombyx silk, Cactoblastis cactus biocontrol. Pests: Helicoverpa armigera cotton bollworm (resistance; Bt cotton), Spodoptera frugiperda fall armyworm, Lymantria dispar defoliation/ballooning; clothes/stored-food moths.

DIPTERA (~4)

1 functional wing pair; hindwings = gyroscopic halteres. Sponging/sucking mouthparts; labellum, pseudotracheae; no true adult ovipositor (oviscapt). Larvae lack jointed thoracic legs. Ecology repeatedly evolves: aquatic, saprophagy, herbivory, predation, parasitoidism, blood-feeding.

3 radiations: lower Diptera (often aquatic, long antennae); Brachycera (robust, shortened antennae, internalised larval head); Schizophora (ptilinum inflates to escape puparium, leaves ptilinal suture). Puparium = hardened final larval skin. Antennae filiform–stylate–aristate; venation/instars often reduced.

Prac ID: venation/costal breaks/closed cells, antennae, chaetotaxy (bristle arrangement). Frons between eyes; ocellar triangle; vibrissae near mouth; holoptic eyes meet, dichoptic separated. Calypters wing-base lobes.

Culicidae scaled wing veins + long proboscis; Tipulidae V-shaped mesonotal suture; Chironomidae non-biting midges; Asilidae facial mystax, predators; Syrphidae spurious vein, pollinators; Bombyliidae bee-mimics, parasitoids; Phoridae crowded anterior veins, humpbacked.

Hematophagy = blood feeding; myiasis = fly larvae infest living vertebrate tissues. Mosquitoes vector malaria/viruses; tephritids fruit pests; agromyzids leaf-miners. Benefits: decomposition, pollination (cacao midges), biocontrol, maggot therapy, forensic evidence; chironomids water-quality indicators. Hippoboscoidea: blood-feeding adults, milk-fed internal larva.

OTHER INVERTEBRATES (~3)

Chelicerates: chelicerae + pedipalps, no antennae; prosoma/opisthosoma. Arachnids usually 4 walking-leg pairs. Acari fused body, larvae usually 3 pairs; ticks barbed hypostome, blood-feeding/disease/paralysis; mites crop pests, allergens or biocontrol.

Spiders: spinnerets, male palpal sperm transfer; mygalomorph fangs parallel, araneomorph opposed. Scorpions: pincer palps + tail sting; pseudoscorpions no tail, phoresy (transport). Horseshoe crabs: book gills, hemocyanin blood, endotoxin detection.

Onychophora: unjointed lobopods, hydrostatic skeleton; arthropod relatives, not annelid “missing links”; slime immobilises prey; desiccation restricts to moist logs/soil; egg-laying to live-bearing. Tardigrada: 4 lobopod pairs, cryptobiotic tun. Cryptobiosis = near-suspended metabolism; eutely = fixed somatic cell number, growth by enlargement.

Nematodes: unsegmented, pseudocoelom, collagenous moulting cuticle; longitudinal muscles against hydrostatic pressure cause thrashing. Molecular evidence + moulting unite Ecdysozoa.

FORENSICS (~2)

Insects in legal investigations. Minimum post-mortem interval (PMI) from oldest insects/development + ecological succession; colonisation may lag death. Fresh–bloat–active decay–post-decay–skeletonisation: early blowflies, then diverse flies/predatory beetles, later beetles dominate.

Blowfly instars: 1/2/3 posterior spiracular slits. Need species ID + temperature history + stage/size; rates species-specific. Rain/cold/access barriers delay colonisation; maggot masses heat; drugs alter growth; light permits night laying. DNA helps ID. Also body movement, drugs, neglect, food contamination.